

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Designed and implemented **soil mechanics** analysis workflows using Python and SQL, reducing manual data processing time by **30%** for geotechnical site assessments.
- Managed project timelines and quality control for **tailings management facility** design reports, ensuring adherence to international design guidelines and delivering 5+ key project documents on schedule.
- Developed technical specifications and cost estimates for **waste rock dump design**, contributing to proposals that secured **£1.5M** in new project awards.
- Provided specialized technical input on **heap leach facility** design and **geosynthetic materials engineering**, collaborating with international teams to solve complex civil engineering challenges.
- Automated the generation of **bill of quantities** and technical drawings using Python scripting, accelerating the design review cycle by **20%**.
- Supported client relationships by providing clear technical documentation and on-site support for **water impoundment** projects, achieving a **95%** client satisfaction rating.

SKILLS

Geotechnical Engineering – Soil Mechanics, Tailings Management Facilities, Heap Leach Facilities, Water Impoundments, Waste Rock Dump Design, Geosynthetic Materials Engineering, Geotechnical Structures

Civil Engineering – Detail Design, Conceptual Design, Construction Support, Site Assessments, Project Management, Quality Control

Software & Scripting – Python, SQL (PostgreSQL, MySQL, Oracle), Shell Scripting, Data Analysis

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Git, Jenkins

Reporting & Documentation – Technical Reports, Drawings, Specifications, Bill of Quantities, Cost Estimates, Confluence

Project Delivery – Time Management, Budget Control, Client Relationship Management, Business Development Support, Mentoring

EDUCATION AND TRAINING

MASTER OF SCIENCE in Data Analytics – National College of Ireland, Dublin

02/2026

BACHELOR OF ENGINEERING in Computer Science – SNS College of Technology, Coimbatore, India

01/2021

PROJECTS

Automated Geotechnical Design Analysis Platform

- Developed a Python-based platform integrating **soil mechanics** principles with client-specific design parameters for **tailings management facilities**.

- Automated the generation of detailed technical reports and **bill of quantities**, reducing preparation time by **40%** and improving accuracy.
- Implemented data validation scripts using **SQL** against PostgreSQL databases to ensure integrity of site assessment data and design inputs, supporting **10+** international projects.

Financial Services Transaction Monitoring Dashboard (HCL Technologies)

- Enhanced a **payments operations dashboard** for a financial services client, focusing on improving transaction monitoring and operational alerting for payment message flows.
- Developed Python scripts for automated data extraction and analysis from production logs (Splunk), reducing incident triage time by **25%**.
- Collaborated with business analysts to refine reconciliation workflows and audit trail reporting, ensuring adherence to GDPR-aware production environment standards.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)